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Blockchain instrument for transferable equity

Abstract

Systems and methods for offering and purchasing tokenized securities on a blockchain platform meeting current and future federal, state, and offering and holding entity rules and regulations. Tokenized securities purchased during or after the tokenized securities offering are tradable on a secondary market. The server computer of the tokenized securities provides an automated transfer capability for tokenized securities holders.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6330547	12/2000	Martin	N/A	N/A
7472098	12/2007	Shields et al.	N/A	N/A
8078490	12/2010	Wallman	N/A	N/A
8504483	12/2012	Foley et al.	N/A	N/A
9692738	12/2016	Wenneman et al.	N/A	N/A
9704143	12/2016	Walker et al.	N/A	N/A
9892460	12/2017	Winklevoss et al.	N/A	N/A
9898782	12/2017	Winklevoss et al.	N/A	N/A
9965804	12/2017	Winklevoss et al.	N/A	N/A
9965805	12/2017	Winklevoss et al.	N/A	N/A
10002389	12/2017	Winklevoss et al.	N/A	N/A
10255108	12/2018	Dillenberger	N/A	G06F 16/28
10255635	12/2018	Winklevoss et al.	N/A	N/A
10333706	12/2018	Smith et al.	N/A	N/A
10339603	12/2018	Sandulli	N/A	N/A
10356094	12/2018	Anton et al.	N/A	N/A
10636180	12/2019	Brogger	N/A	N/A
2002/0002520	12/2001	Gatto	705/36R	G06Q 40/04
2003/0149646	12/2002	Chen	705/35	G06Q 40/00
2005/0091133	12/2004	Ballman	705/35	G06Q 40/00
2007/0078745	12/2006	Kalt	N/A	N/A
2007/0233589	12/2006	Vasinkevich	N/A	N/A
2010/0325062	12/2009	O'Shaughnessy et al.	N/A	N/A
2011/0106685	12/2010	Silbert	N/A	N/A
2012/0011044	12/2011	Vasinkevich	N/A	N/A
2012/0173403	12/2011	Foley	705/37	G06Q 99/00
2014/0012780	12/2013	Sanders	705/36R	G06Q 30/0213
2014/0108277	12/2013	Dresner et al.	N/A	N/A

2015/0278168	12/2014	Hawa et al.	N/A	N/A
2015/0379510	12/2014	Smith	N/A	N/A
2016/0012465	12/2015	Sharp	N/A	N/A
2016/0035025	12/2015	Lagman	N/A	N/A
2016/0098723	12/2015	Feeney	N/A	N/A
2016/0321751	12/2015	Creighton, IV et al.	N/A	N/A
2016/0344737	12/2015	Anton et al.	N/A	N/A
2017/0011460	12/2016	Molinari	N/A	G06F 21/645
2017/0017779	12/2016	Huang et al.	N/A	N/A
2017/0085545	12/2016	Lohe et al.	N/A	N/A
2017/0103391	12/2016	Wilson, Jr. et al.	N/A	N/A
2017/0155515	12/2016	Androulaki et al.	N/A	N/A
2017/0221052	12/2016	Sheng et al.	N/A	N/A
2017/0228822	12/2016	Creighton, IV et al.	N/A	N/A
2017/0289111	12/2016	Voell et al.	N/A	N/A
2017/0308893	12/2016	Saraniacki	N/A	N/A
2017/0330174	12/2016	DeMarinis et al.	N/A	N/A
2018/0096349	12/2017	McDonald et al.	N/A	N/A
2018/0294967	12/2017	Roberts et al.	N/A	N/A
2018/0323963	12/2017	Stollman	N/A	N/A
2019/0005469	12/2018	Dhupkar	N/A	G06Q 20/06
2019/0035014	12/2018	Bell et al.	N/A	N/A
2019/0043048	12/2018	Wright et al.	N/A	N/A
2019/0057448	12/2018	Sandulli	N/A	N/A
2019/0066206	12/2018	Marks	N/A	N/A
2019/0080402	12/2018	Molinari et al.	N/A	N/A
2019/0080404	12/2018	Molinari et al.	N/A	N/A
2019/0080405	12/2018	Molinari et al.	N/A	N/A
2019/0080406	12/2018	Molinari et al.	N/A	N/A
2019/0080407	12/2018	Molinari et al.	N/A	N/A
2019/0102409	12/2018	Shi et al.	N/A	N/A
2019/0130483	12/2018	Jong	N/A	N/A
2019/0139136	12/2018	Molinari et al.	N/A	N/A
2019/0156304	12/2018	Hudson	N/A	H04L 9/0637
2019/0164151	12/2018	Doney et al.	N/A	N/A
2019/0180372	12/2018	Creighton, IV et al.	N/A	N/A
2019/0180373	12/2018	Creighton, IV et al.	N/A	N/A
2019/0188793	12/2018	Molinari et al.	N/A	N/A
2019/0190697	12/2018	Cunico et al.	N/A	N/A
2019/0197622	12/2018	Molinari et al.	N/A	N/A
2019/0251629	12/2018	Gordon, III et al.	N/A	N/A
2019/0333051	12/2018	Brogger	N/A	N/A
2019/0340689	12/2018	Gordon, III et al.	N/A	N/A

2019/0347660	12/2018	Wilkinson et al.	N/A	N/A
2019/0385155	12/2018	Silverman	N/A	G06Q 40/04
2020/0076576	12/2019	Ahlbäck et al.	N/A	N/A
2020/0160320	12/2019	Williams	N/A	G06Q 20/36
2020/0175623	12/2019	Howie	N/A	N/A
2020/0265518	12/2019	Spangenberg et al.	N/A	N/A
2020/0320631	12/2019	Gordon, III et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2560671	12/2017	GB	H04W 12/06
2560671	12/2018	GB	N/A
2019109930	12/2019	JP	N/A
2017090041	12/2016	WO	N/A
2017197110	12/2016	WO	N/A
2017213943	12/2016	WO	N/A
WO-2019067603	12/2018	WO	G06Q 20/3678

OTHER PUBLICATIONS

Bitcoinist.net “Get Ready to Be Surprised During the Additional Round of e-Chat ICO” Jan. 1, 2018
Bitcoin Insider, Retrieved from the Internet: <https://www.bitcoininsider.org/article/13605/get-ready-be-surprised-during-additional-round-e-chat-ico> (Year: 2018). cited by examiner

Amano et al. “The Regulated Token (R-Token) Standard”, Feb. 2018, Harbor, Retrieved from the Internet: <https://harbor.com/rtokenwhitepaper.com> (Year: 2018). cited by applicant

Aste et al., Blockchain Technologies: The Foreseeable Impact on Society and Industry, Published in: Computer (vol. 50, Issue: 9, pp. 18-28), Sep. 22, 2017
(<https://ieeexplore.ieee.org/document/8048633>) (Year: 2017). cited by applicant

Bitcoinist.net “Get Ready to Be Surprised During the Additional Round of e-Chat ICO”, Jan. 1, 2018, Bitcoin Insider, Retrieved from the Internet: <https://www.bitcoininsider.org/article/13605/get-ready-be-surprised-during-additional-round-e-chat-ico> (Year: 2018). cited by applicant

Euler, The Token Classification Framework: A multi-Dimensional tool for understanding and classifying crypto tokens, Published—Untitled INC, Jan. 1, 2018 (<http://www.untitled-inc.com/the-token-classification-framework-a-multi-dimensional-tool-for-understanding-and-classifying-crypto-tokens/>) (Year: 2018). cited by applicant

Stark, Law 360: ICO Token Purchasers May Be Entitled To A Refund, Published Portfolio Media. Inc.: Nov. 13, 2017 (<https://www.law360.com/articles/985189/ico-token-purchasers-may-be-entitled-to-a-refund>) (Year: 2017). cited by applicant

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS (1) The present invention is related to and claims priority from the following U.S. patent documents: this application is a continuation of U.S. patent application Ser. No. 18/411,734, filed Jan. 12, 2024, which is a continuation of U.S. patent application Ser. No. 18/215,435, filed Jun. 28, 2023, which is a continuation of U.S. patent application Ser. No. 17/900,395, filed Aug. 31, 2022 and issued as U.S. Pat. No. 11,694,264, which is a continuation of U.S. patent application Ser. No. 17/719,951, filed Apr. 13, 2022 and issued as U.S. Pat. No. 11,436,679, which is a continuation of U.S. patent application Ser. No. 17/402,105, filed Aug. 13, 2021 and issued as U.S. Pat. No. 11,308,559, which is a continuation of U.S. patent application Ser. No. 16/926,050, filed Jul. 10, 2020 and issued as U.S. Pat. No. 11,094,014, which is a continuation of U.S. patent application Ser. No. 16/271,447, filed Feb. 8, 2019 and issued as U.S. Pat. No. 10,713,722, which claims priority from U.S. Provisional Patent Application No. 62/630,559, filed Feb. 14, 2018, each of which is incorporated herein by reference in its entirety.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) This invention relates to systems and methods for offering, purchasing, and reselling blockchain instruments for transferable equity. More particularly, the present invention provides systems and methods for offering, purchasing, and reselling tokenized securities on a blockchain platform.

2. Description of the Prior Art

(2) A blockchain is a distributed database storing a registry of transactions and records across a peer-to-peer network. The registry is replicated on every computer that uses the network. The transactions and records are built into blocks and secured through cryptography. Each block contains a timestamp and a hash link to a previous block. A cryptocurrency is a digital medium of exchange using cryptography to secure the transactions and to control the creation of additional units of the cryptocurrency. The blockchain technology is the underlying technology for the first cryptocurrency Bitcoin which was created in 2009. Many different cryptocurrencies have been created since then. An initial coin offering (ICO) has become a wildly popular means of crowdfunding by launching a new cryptocurrency. During the ICO, a company offers a new cryptocurrency or token which can be used for products or services on their platform in the future in exchange for cryptocurrencies of immediate, liquid value, such as Bitcoin and ETHEREUM. ICOs have provided a means by which start-up companies are able to avoid costs of regulatory compliance and intermediary financial organizations, while increasing risk for investors. ICOs often fall outside existing regulations or need to be regulated depending on the nature of the project. Some jurisdictions, such as China and South Korea, have banned ICOs altogether. In the United States, the Securities and Exchange Commission (SEC) has been sending out warnings regarding ICOs and cryptocurrencies and indicating tokens offered during an ICO may be considered securities.

(3) Exemplary US Patent Documents relevant to the prior art include:

(4) U.S. Pat. No. 9,704,143 for Cryptographic currency for securities settlement by inventors Walker et al., filed Oct. 30, 2014 and issued Jul. 11, 2017, describes security settlement in financial markets and cryptographic currencies. Particular portions of the disclosure are directed to a cryptographic currency protocol and to a cryptographic currency that includes a positional item. The cryptographic currency protocol supports a virtual wallet that, in various embodiments, is a security and cash account for storing and managing the cryptographic currency. Opening a transaction via the virtual wallet to transfer the cryptographic currency is a strong guarantee of the availability of funds in the virtual wallet because, e.g., funds are not transacted unless the commit phase is successful.

(5) U.S. Patent Publication No. 2016/0012465 for System and method for distributing, receiving, and using funds or credits and apparatus thereof by inventor Sharp, filed Feb. 8, 2015 and published Jan. 14, 2016, describes a system for performing various methods of sending, receiving,

distributing, and utilizing funds and/or credits. In many embodiments, various communications platforms and/or protocols may be employed. Methods of sending funds or credits may be practiced in different environments, including physical and electronic environments. According to some preferred embodiments, users may perform a variety of transactions including various gifting functions, re-gifting functions, and social interactions simply, through various types of electronic communications, including, but not limited to electronic messaging.

(6) U.S. Patent Publication No. 2017/0308893 for Asset and obligation management using flexible settlement times by inventor Saraniecki, filed Aug. 25, 2016 and published Oct. 26, 2017, describes a system and method for managing a transaction having at least one enduring obligation and at least one repo obligation with respect to a plurality of assets, where the system includes at least one signing server for authorizing the at least one enduring obligation and the at least one repo obligation.

(7) U.S. Patent Publication No. 2017/0085545 for Smart Rules and Social Aggregating, Fractionally Efficient Transfer Guidance, Conditional Triggered Transaction, Datastructures, Apparatuses, Methods and Systems by inventors Lohe et al., filed Jul. 14, 2016 and published Mar. 23, 2017, describes the Smart Rules and Social Aggregating, Fractionally Efficient Transfer Guidance, Conditional Triggered Transaction, Datastructures, Apparatuses, Methods and Systems (“SOCOACT”) transforming smart contract request, crypto currency deposit request, crypto collateral deposit request, crypto currency transfer request, crypto collateral transfer request inputs via SOCOACT components into transaction confirmation outputs. A selection of a crypto smart rule type for a crypto smart rule associated with an aggregated crypto transaction trigger entry may be obtained from a user. A crypto smart rule generator user interface (UI) for the selected crypto smart rule type may be provided. Selections of a threshold constraint and of an aggregated blockchain oracle for the crypto smart rule may be obtained from the user via the UI. The aggregated crypto transaction trigger entry may be generate based on the selections and instantiated in a socially aggregated blockchain datastructure.

(8) U.S. Patent Publication No. 2017/0221052 for Computationally Efficient Transfer Processing and Auditing Apparatuses, Methods and Systems by inventors Sheng et al., filed Apr. 12, 2017 and published Aug. 3, 2017, describes the Computationally Efficient Transfer Processing, Auditing, and Search Apparatuses, Methods and Systems (“SOCOACT”) transforming smart contract request, crypto currency deposit request, crypto collateral deposit request, crypto currency transfer request, crypto collateral transfer request inputs via SOCOACT components into transaction confirmation outputs. Also, SOCOACT transforms transaction record inputs via SOCOACT components into matrix and list tuple outputs for computationally efficient auditing. A blockchain transaction data auditing apparatus comprises a blockchain recordation component, a matrix Conversion component, and a bloom filter component. The blockchain recordation component receives a plurality of transaction records for each of a plurality of transactions, each transaction record comprising a source address, a destination address, a transaction amount and a timestamp of a transaction; the source address comprising a source wallet address corresponding to a source digital wallet, and the destination address comprising a destination wallet address corresponding to a destination virtual currency wallet; verifies that the transaction amount is available in the source virtual currency wallet; and when the transaction amount is available, cryptographically records the transaction in a blockchain comprising a plurality of hashes of transaction records. The Bloom Filter component receives the source address and the destination address, hashes the source address using a Bloom Filter to generate a source wallet address, and hashes the destination address using the Bloom Filter to generate a destination wallet address. The Matrix Conversion component adds the source wallet address as a first row and a column entry to a stored distance matrix representing the plurality of transactions, adds the destination wallet address as a second row and column entry to the stored distance matrix representing the plurality of transactions, adds the transaction amount and the timestamp as an entry to the row corresponding to the source wallet address and the column

corresponding to the destination wallet address; and generate a list representation of the matrix, where each entry in the list comprises a tuple having the source wallet address, the destination wallet address, the transaction amount and the timestamp.

(9) U.S. Patent Publication No. 2017/0103391 for Digital asset intermediary electronic settlement platform by inventors Wilson et al., filed Dec. 22, 2016 and published Apr. 13, 2017, describes a digital asset settlement method including receiving from a first user an authorization for a conditional transaction involving a digital right, which has been digitized on a distributed ledger, matching the authorization for transaction from the first user with an authorization for transaction from at least one other user, settling the transaction between at least the first and other users if the conditional is met, and memorializing the settled transaction on the distributed ledger.

SUMMARY OF THE INVENTION

(10) In one embodiment of the present invention, systems and methods for offering and purchasing tokenized securities on a blockchain platform are provided. At least one user device for at least one investor is configured and constructed in network communication with a server computer of a tokenized securities offering entity. The at least one user device transmits user input data to the server computer of the tokenized securities offering entity via a graphical user interface (GUI). The server computer of the tokenized securities offering entity transmits a link to at least one accreditation agency to the at least one user device for investor accreditation. The server computer of the tokenized securities offering entity accesses to and synchronizes with at least one database of the at least one accreditation agency in real time, and creates an up-to-date whitelist of accredited investors based on accreditation information obtained from the at least one database of the at least one accreditation agency. The server computer of the tokenized securities offering entity verifies the accreditation status of the at least one investor and sends a link to at least one tokenized securities contract deployed on the blockchain platform. The at least one user device sends an acceptance message after the at least one investor review documents included in the at least one tokenized securities contract on the blockchain platform. The at least one user device transmits a predetermined amount of cryptocurrency from a uniquely identified account or cryptocurrency account or a digital wallet of the at least one investor to an escrow account of the tokenized securities offering entity on the blockchain platform. In one embodiment, the at least one tokenized securities contract has an escrow function. At least one security token is sent to the uniquely identified account or cryptocurrency account or the digital wallet of the at least one investor, and the predetermined amount of the currency is sent to a uniquely identified account or a digital wallet of the tokenized securities offering entity.

(11) In another embodiment, the present invention provides systems and method for selling (or reselling) tokenized securities on the blockchain platform. Tokenized securities purchased during the tokenized securities offering (or through existing manual means) are tradable on a secondary market usually after a rest period. In another embodiment, the server computer of the tokenized securities offering entity provides a token transfer system for selling (or reselling) security tokens issued by the tokenized securities offering or holding entity.

(12) These and other aspects of the present invention will become apparent to those skilled in the art after a reading of the following description of the preferred embodiment when considered with the drawings, as they support the claimed invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates a system of purchasing security tokens according to one embodiment of the present invention.

(2) FIG. 2 is a timeline throughout a life cycle of BITE tokens according to one embodiment of the

present invention.

(3) FIG. 3 illustrates interactions between different entities during a security token offering according to one embodiment of the present invention.

(4) FIG. 4 illustrates a token purchasing process on blockchain during a security token offering according to one embodiment of the present invention.

(5) FIG. 5 is a timeline throughout a life cycle of tokenized securities according to one embodiment of the present invention.

(6) FIG. 6 illustrates transactions and contracts recorded in blocks on the ETHEREUM blockchain according to one embodiment of the present invention

(7) FIG. 7 illustrates transactions for taken contract between an investor and a security token offering entity on ETHEREUM blockchain according to one embodiment of the present invention.

(8) FIG. 8 lists investor documents in a token purchase process according to one embodiment of the present invention.

(9) FIG. 9 is a tokenized securities offering timeline according to one embodiment of the present invention.

(10) FIG. 10 is a flowchart of a token purchase process according to one embodiment of the present invention.

(11) FIG. 11 illustrates a proxy contract according to one embodiment of the present invention.

(12) FIG. 12 illustrates an escrow contract according to one embodiment of the present invention.

(13) FIG. 13 illustrates an aggregate contract according to one embodiment of the present invention.

(14) FIG. 14 illustrates a typical contract according to one embodiment of the present invention.

(15) FIG. 15 is a diagram illustrating a signup process for offering tokenized securities on a blockchain platform according to one embodiment of the present invention.

(16) FIG. 16 is a diagram illustrating a trading process for a tokenized securities seller on a blockchain platform according to one embodiment of the present invention.

(17) FIG. 17 is a schematic diagram illustrating a cloud-based system of the present invention.

(18) FIG. 18 is another schematic diagram illustrating a cloud-based system of the present invention.

DETAILED DESCRIPTION

(19) The blockchain technology is based on existing communication protocols (e.g., HTTP, RPC), cryptography (grown from public key cryptography in 1976), distributed peer-to-peer sharing mechanisms (e.g., NAPSTER, BITTORRENT), and a distributed set of databases kept in synchronization based on time. The blockchain technology is a technology that permanently records events or transactions on a network in a transparent, auditable, and irrefutable way. A blockchain ledger is stored on each blockchain node participating in or comprising a network. Blockchain nodes include, but are not limited to servers, mobile devices, work stations, or any networked client that can interface with an IP-based network and can operate an operating system capable of processing blocks. Blockchain is a loose specification rather than a specific implementation, which is capable of unlocking monopoly power over information in infrastructure systems for telecommunications, healthcare, finance, energy, and government. Blockchain also disintermediates “middle men” such as broker dealers, banks, transfer agents, or any third party in information or transactions that are utilized for trust in the transmittal of data or the execution of a transaction. In an introduction to blockchain applications in *The Business of Blockchain* by William Mougayar (2016), which is incorporated herein by reference in its entirety, it is established that just as the Web could not exist without the Internet, blockchains could not exist without the Internet, and thus, the use of blockchains within the systems and methods of the present invention provide that it is not merely an abstract idea, since it is inextricably tied to Internet technology.

(20) A smart contract is a computer protocol intended to digitally facilitate, verify, or enforce the negotiation or performance of a contract. In the context of blockchain, smart contracts are self-

executing codes on a blockchain that automatically implements the terms of an agreement between parties. Tokenized securities contracts deployed on a blockchain platform by a tokenized securities offering entity are based on the smart contract technology.

(21) The present invention provides systems and methods for offering, purchasing, and reselling blockchain instruments for transferable equity. In one embodiment, the blockchain instruments for transferable equity are tokenized securities offered and purchased on a blockchain platform. In one embodiment, the blockchain instruments for transferable equity are tokenized securities held by an entity (“holder entity”) that are available for sale by existing investors to purchasers on a blockchain platform. In one embodiment, the blockchain instruments for transferrable equity are tokenized securities, and each security token is unique. Advantageously, the systems and methods in the present invention disintermediate brokers in buying and selling securities and other financial instruments.

(22) In one embodiment of the present invention, systems and methods for offering, purchasing, and reselling tokenized securities on a blockchain platform are provided. At least one user device for at least one investor is in network communication with at least one server computer operable for use by a tokenized securities offering entity. The at least one user device transmits user input data to the at least one server computer operable for use by the tokenized securities offering entity via a graphical user interface (GUI). The at least one server computer operable for use by the tokenized securities offering entity transmits a link to at least one accreditation agency to the at least one user device for investor accreditation. The at least one server computer operable for use by the tokenized securities offering entity accesses and synchronizes with at least one database of the at least one accreditation agency in real time, and creates an up-to-date whitelist of accredited investors based on accreditation information obtained from the at least one database of the at least one accreditation agency. The at least one server computer operable for use by the tokenized securities offering entity verifies the accreditation status of the at least one investor and sends a link to at least one tokenized securities contract deployed on the blockchain platform. The at least one user device sends an acceptance message after the at least one investor reviews documents included in the at least one tokenized securities contract on the blockchain platform. The at least one user device transmits a predetermined amount of currency from a uniquely identified account or a digital wallet of the at least one investor to an escrow account of the tokenized securities offering entity on the blockchain platform. In one embodiment, the at least one tokenized securities contract has an escrow function. At least one security token is sent to the uniquely identified account or the digital wallet of the at least one investor, and the predetermined amount of the currency is sent to a uniquely identified account or a digital wallet of the tokenized securities offering entity.

(23) In one embodiment of the present invention, systems and methods for purchasing tokenized securities are provided. At least one server computer operable for use by the tokenized securities offering (or holding) entity receives input data from an investor for registration via a GUI on a user device. An offering entity is one selling securities to raise capital, a holding entity is one that has already used the securities to raise capital, who now “holds” the tokenized securities. The input data includes a legal name and an e-mail address and other required information. In one embodiment, other required information further includes investor accreditation status. In one embodiment, the at least one server computer operable for use by the tokenized securities offering entity provides a third-party agency link for investor accreditation via the GUI. Investor accreditation lasts a certain period of time and then expires. Renewal is needed in order to maintain accredited investor status. The at least one server computer operable for use by the tokenized securities offering entity is operable to access and synchronize with various data sources for investor accreditation information automatically. The at least one server computer operable for use by the tokenized securities offering entity creates and updates a whitelist of accredited investors based on the investor accreditation information from various data sources. The at least one server computer operable for use by the tokenized securities offering entity verifies the investor

accreditation status of a registered user based on the up-to-date whitelist. If the investor accreditation status is included in the whitelist, the at least one server computer operable for use by the tokenized securities offering entity transmits a link to at least one tokenized securities contract including various documents to the user device for review. The at least one tokenized securities contract is deployed on a blockchain platform. In one embodiment, the link is sent to a cryptocurrency account or a digital wallet of the investor accessible by the user device. The user device transmits an acceptance message indicating the investor accepts the terms and conditions in the various documents included in the at least one tokenized securities contract to the blockchain platform. The user device is then operable to purchase at least one security token via the cryptocurrency account or digital wallet of the investor. Thus, a purchase order is complete and the at least one tokenized securities contract is let. In one embodiment, tokenized securities are at rest for a certain period of time before being traded in order to forbid broker-dealer behaviors. A rest period is determined by a governing entity regarding securities law during which public or private trading of securities are prohibited. For example, SEC rule 144 specifies that “if the company that issued the securities is a “reporting company” in that it is subject to the reporting requirements of the Securities Exchange Act of 1934, then invested investors must hold the securities for at least six months. If the issuer of the securities is not subject to the reporting requirements, then invested investors must hold the securities for at least one year. In one embodiment, the at least one server computer operable for use by the tokenized securities offering entity is operable to burn security tokens, which are tokenized securities, left after the offering period. In another embodiment, the sever computer of the tokenized securities offering entity provides a marketplace where tokenized securities contracts are traded on the blockchain platform. The at least one server computer operable for use by the at least one tokenized securities offering entity includes one or more server computers, which is standalone or network-based or cloud-based.

(24) The description below provides details for steps of registering, verifying, reviewing, accepting, and investing according to one embodiment of a method of purchasing tokenized securities during a tokenized securities offering. In another embodiment, the order of the above-mentioned steps varies. For example, but not for limitation, the verification step is after the invest step.

(25) Register

(26) In one embodiment, participants in tokenized securities offerings or purchasing of existing tokenized securities include individuals, entities, joint tenancy, brokers, dealers, trusts and other legally recognized forms of organization or regulatory status. At the registering stage, information required from an investor by the server computer of the tokenized securities offering entity includes legal name, country of residence, and email address. A privacy notice is sent to the investor via a user device. At least one uniquely identified account is required for the investor to participate in the tokenized securities offering.

(27) In one embodiment, the at least one uniquely identified account is a hash value generated by a hash algorithm in the blockchain. In one embodiment, the at least one uniquely identified account is a blockchain account, a cryptocurrency account, or a digital wallet. In one embodiment, the at least one uniquely identified account is generated by other mathematical algorithm (e.g., quantum mechanics).

(28) In one embodiment, investor accreditation is required to participate in the tokenized securities offering. In one embodiment, accredited investor is defined in 17 CFR 230.501 (a). For example, accredited investors include certain institutional investors, private business development companies, entities with total assets in excess of \$5,000,000, insiders, individuals with high net worth, individuals with high income, trusts having total assets in excess of \$5,000,000, and entities owned entirely by accredited investors, according to specific rules in 17 CFR 230.501 (a). In one embodiment, a certificate ID for investor accreditation is required by the server computer for registration. If the certification ID is not provided, the server computer of the tokenized securities offering entity sends a link to at least one third-party accreditation agency (e.g.,

VerifyInvestor.com, EarlyIQ.com, SeedInvest.com) to the user device via e-mail, text message, a social media account, or any other communication method, preferably with two-step authentication for user accreditation. The user device gives the server permission to access the investor accreditation information from the at least one third-party accreditation agency. In another embodiment, the registered user is enabled for self-accreditation by providing qualification documents, and the server computer of the tokenized securities offering entity is operable to review the qualification documents and grant permission to the registered user for participating in the tokenized securities offering if the registered user is qualified as an accredited investor.

(29) Under the federal securities laws in the US, a company or private fund is not permitted to offer or sell securities unless the transaction has been registered with the Securities and Exchange Commission (SEC) or an exemption from registration is available. Certain securities offerings that are exempt from registration are only offered to, or purchased by, persons who are accredited investors.

(30) In another embodiment, an investor is represented by a dealer or broker, and as such the dealer or broker will additionally provide their registration credentials such as their Financial Industry Regulatory Authority (FINRA) Central Registration Depository (CRD) numbers and SEC numbers.

(31) Verify

(32) The server computer of the tokenized securities offering or holding entity is operable to access and synchronize with various data sources for investor accreditation information. Investor accreditation status lasts a certain period of time and renewal is needed to keep the investor accreditation status alive. The server computer of the tokenized securities offering entity automatically creates and maintains an up to date whitelist of accredited investors based on the investor accreditation information obtained from various data sources. The server computer of the tokenized securities offering entity is operable to add self-accredited investors to the whitelist if they pass a review process conducted by the tokenized securities offering entity. The whitelist of accredited investors includes legal names, social security numbers (SSNs), tax identification numbers (TINs), uniquely identified accounts (e.g., cryptocurrency accounts or digital wallet addresses, fiat currency bank account information), accreditation expiration dates, and other information related to accredited investors. In one embodiment, the tokenized securities offering entity also creates and maintains a blacklist and blocks domain names, individuals, competitors and others with malicious intent from the token purchase or exchange.

(33) The server computer of the tokenized securities offering entity compares the certificate ID and its issuing agency for the registered user with the up-to-date whitelist and blacklist of accredited investors for verification. Once the accreditation status of the registered user is verified, the server computer of the tokenized securities offering entity transmits a link to at least one tokenized securities contract including various document to a cryptocurrency account or digital wallet of the registered user accessible to the user device for review. In one embodiment, tokenized securities contracts are deployed on the ETHEREUM blockchain. An ETHEREUM account or an ETHEREUM wallet is required for purchasing security tokens in the tokenized securities offering. In another embodiment, other blockchain platforms and/or other cryptocurrencies are used for offering and purchasing tokenized securities. In another embodiment, the tokenized securities contracts are deployed on another blockchain platform. The blockchain platforms for the tokenized securities contracts can be on a private and/or a public network.

(34) To confirm investor accreditation, the server computer of the tokenized securities offering entity is operable to automatically access records related to accreditation in various databases, retrieve accreditation information automatically, and/or inquire financial institutions for investor accreditation status. In one embodiment, the tokenized securities offering entity requires at least \$1 million worth of Ether or any generally accepted cryptocurrencies or fiat currencies such as U.S. dollars in order to confirm accreditation. SEC regulations currently require accreditation based on U.S. dollars.

(35) In one embodiment, if the investor is represented by a dealer or broker, the server computer of the tokenized securities offering entity is operable to use a blockchain oracle to confirm the dealer or broker's registration credentials and automatically whitelist the dealer or broker, if such representation is allowed in the offering or resale of tokenized securities.

(36) In one embodiment, all accreditation information is recorded on various data sources over the Internet. The various data sources include but are not limited to databases for accreditation service agencies and blockchains recording accreditations and other data feeds requiring investor accreditation. In one embodiment, a blockchain oracle is applied to automatically retrieve accreditation information in real time from various data sources. An oracle, in the context of blockchains and smart contracts, is an agent that finds and verifies real world occurrences and submits this information to a blockchain to be used by smart contracts. This agent can be software, hardware, or human. In one embodiment of the present invention, the oracle is a software-based oracle and programmed to search for and parse text for accreditation information.

(37) The server computer of the tokenized securities offering entity utilizes at least one blockchain oracle to retrieve accreditation information. Different data sources work differently for investor accreditation. Information retrieved from various data sources are normalized before the information is included in the whitelist. The server computer of the tokenized securities offering entity is operable to confirm the accreditation status of a registered user automatically based on the up-to-date whitelist of accredited investors.

(38) Review

(39) Once a party has been allowed to view the disclosure documents, the offering or holding entity manage which documents are mandatory to be reviewed and which are not. The various disclosure documents in the at least one tokenized securities contract include mandatory and non-mandatory documents. For example, but not for limitation, mandatory documents include Private Placement Memorandums (PPM) (including business plans, risks, and financial information), Contract of Securities Offering, Pricing Adjustments, and Purchase Agreements; non-mandatory documents include Company Bylaws, Contracts, Licensing Agreements, and Whitepapers. In one embodiment, mandatory documents are required to be downloaded. Mandatory documents include watermarks and other security features that the server computer of the tokenized securities offering entity requires to be reported back via the blockchain platform for automatic confirmation that the mandatory documents have been downloaded, which is an automated indication that the mandatory documents have been reviewed by the registered user. Alternatively, an input from the registered user is received by the blockchain platform indicating that the mandatory documents have been reviewed. In one embodiment, the disclosure documents are disclosed only to parties as desired by the offering entity or holding entity.

(40) Documents reviewed by the registered user are hashed based on a secure hash algorithm (e.g., SHA-3) and each hash is stored in the blockchain. Hashes of the reviewed documents uniquely represent what the documents are. If a document is known, then the hash of it is known based on the secure hashing algorithm used. In one embodiment, the server computer of the tokenized securities offering or holding entity is operable to compare a hash representing a reviewed document by a registered user to the hashes of the documents provided by the tokenized securities offering entity to confirm that the registered user has reviewed the mandatory documents.

(41) Accept

(42) User information is required for SEC compliance from the registered user and recorded on the blockchain platform. For individuals, required information includes legal name, address, and SSN. For entities, required information includes company name, address, and TID. Other information for the registered user recorded on the blockchain include a uniquely identified account on a blockchain platform (e.g., a cryptocurrency account or a digital wallet address), automated clearing house account, bank account (e.g., source or receipt of payment information), accreditation expiration date, and the hash of reviewed documents.

(43) After reviewing various documents in the at least one tokenized securities contract, the user device transmits an acceptance message regarding the terms and conditions in the tokenized securities offering. In one embodiment, the terms include not being a bad actor according to the Office of Foreign Assets Control (OFAC), not violating the securities laws, not being a broker or dealer, not intending to purchase to resell or any other broker or dealer activity, etc. The terms also include representations and warranties, private placement memorandum (PPM), anti-money laundering (AML), Know Your Customer (KYC). The AML and KYC terms can also be verified via third-party reporting agencies such as banks, US Treasury, etc. via a software oracle.

(44) Purchase

(45) At purchase stage, a purchase order is completed by the registered user paying a certain amount of cryptocurrency or fiat currency for at least one security token specified in at least one tokenized securities contract. In one embodiment, the certain amount of currency is transferred from the unique identified account or digital wallet of the registered user to an escrow account for the tokenized securities offering entity on the blockchain platform. In one embodiment, the escrow account is a third-party escrow agency on the blockchain platform. In another embodiment, the escrow account is a third-party escrow agency in a banking system for fiat currency. In another embodiment, the tokenized securities contracts deployed on the blockchain platform provide an escrow function by holding cryptocurrencies or banking system representations of fiat currency, and security tokens until a predetermined threshold is reached. The predetermined threshold can be a date, a number of investors invested, a minimum raise, or other metrics determined by the tokenized securities offering entity. After the predetermined threshold is reached, the certain amount of cryptocurrency paid by the registered user is released to a cryptocurrency account or a digital wallet of the tokenized securities offering entity, and the at least one security token is released to the cryptocurrency account or digital wallet of the registered user corresponding to the at least one tokenized securities contract. Thus, the at least one tokenized securities contract is let and recorded on the blockchain platform. Once the at least one tokenized securities contract is completed, it is immutable.

(46) A purchase agreement is a binding contract between the tokenized securities offering entity and investors. The purchase agreement defines what security is being acquired. During a purchase agreement process, an investor agrees to the terms and conditions in the purchase agreement and makes their representations (e.g., accredited investors, not a bad actor, not a foreign asset holder in restricted countries, not performing an illegal activity, etc.). A purchase order is the transfer of funds in exchange of securities) (e.g., equity, debt, option). During a purchase order process, the tokenized securities offering entity verifies that the investor is not on a blacklist (e.g., competitor, affiliate), passes the OFAC check, and is an accredited investor and that the seller is authorized to sell the security (e.g., not an executive, broker/dealer) by the offering or holding entity. How and when related information is gathered and processed vary in different embodiments of the present invention.

(47) Rights of the Tokenized Securities Offering Entity

(48) In an Initial Public Offering (IPO), a red herring prospectus issued by an issuing company to potential investors contains information associated with the issuing company and their IPO. Information in the red herring prospectus is not complete and is subject to change. Similarly, documents provided in the tokenized securities offering are changeable by the tokenized securities offering entity with SEC compliance during the tokenized securities offering. Once a document is updated, it is a new document and will have a new hash based on the secure hash algorithm used by the tokenized securities offering entity. If a document is updated after reviewed by a registered user but before the purchase process is completed, the server computer of the tokenized securities offering entity redirects the registered user to go back to review the updated document via the GUI in order to complete the purchase process. In one embodiment, the GUI is operable to read and interpret blockchain messages. If a document is updated after a registered user completed a

purchase order during the offering period, the server computer of the tokenized securities offering entity still provides the updated document to the registered user and the registered user is enabled to select to reverse or keep the purchase order. Contracts of tokenized securities purchased at different times during the tokenized securities offering are sometimes different due to document updates.

(49) The server computer of the tokenized securities offering entity is operable to change the registration process, change terms of offering, approve tokenized securities transferring, and make other updates in order to comply with rules and regulations in a certain jurisdiction. In one embodiment, systems and methods for offering and purchasing tokenized securities are compliant with the SEC. The server computer of the tokenized securities offering entity is operable to update the registrations process, update offering terms, and approve security token transferring with SEC compliance.

(50) The server computer of the tokenized securities offering entity has enforcement mechanisms to execute against security tokens issued by the tokenized securities offering entity. In one embodiment, if the tokenized securities offering entity does not raise a predetermined minimum amount of funds (e.g., cryptocurrency and/or fiat currency) of liquid value, all the currency received from investors is reversed back to the investors. In one embodiment, if the tokenized securities offering entity does not sell a predetermined number of tokens, all the currency received from investors is reversed back to the investors. In one embodiment, after the tokens are sold, there is a uniform return price that is based upon a moving average of the value of ETHEREUM. For example, but not for limitation, if ETHEREUM is accepted by the tokenized securities offering entity, the moving average of the value of the ETHEREUM is a 90-day moving average. In one embodiment, if an investor does not comply with the terms and conditions as accepted by the investor, the server computer of the tokenized securities offering entity is operable to reject the noncompliant investor as a holder of the security tokens issued by the tokenized securities offering entity by reversing the transactions with invested users and/or seizing the security tokens purchased by noncompliant investors on the blockchain platform.

(51) The enforcement mechanisms apply to accredited investors throughout an entire trading cycle. In one embodiment, if an investor has lost the accreditation status since renewal is required after a certain period of time, the server computer of the tokenized securities offering entity is operable to request the discredited investor to sell the security tokens. In one embodiment, the security tokens are sold by the discredited investor. In another embodiment, the server computer of the tokenized securities offering entity is operable to sell the security tokens on behalf of the discredited investor on the blockchain platform. Optionally, the discredited investor can still retain the security tokens with regulatory compliance. Section 12(g) of the Exchange Act requires an issuer with total assets of more than \$10 million and a class of securities held of record by either 2,000 persons, or 500 persons who are not accredited investors, to register that class of securities with the Commission. However, securities issued pursuant to Regulation Crowdfunding are conditionally exempted from the record holder count under Section 12(g) if the following conditions are met: (1) the issuer is current in its ongoing annual reports required pursuant to Regulation Crowdfunding; (2) has total assets as of the end of its last fiscal year of \$25 million or less; and (3) has engaged the services of a transfer agent registered with the SEC.

(52) In one embodiment, the systems and methods of the present invention do not allow for buying and selling security tokens at the same time or within a specified time period. The amount of security tokens a registered user is planning to purchase is held in escrow on the blockchain while the registered user is in the process of purchasing, and the security tokens held in escrow cannot be sold by the tokenized securities offering entity to prevent double-spending. Additionally, once a purchase is completed there are rules to be enforced such that the purchaser cannot resell the security tokens for some period of time to comply with SEC rules for 'making a market' (i.e., a rest period).

(53) Token Market for Token Transfer

(54) In one embodiment, the tokenized securities offering entity provides a token market with a token transfer framework facilitating token exchanges after an offering period. In one embodiment, restrictions are introduced to prevent the tokenized securities offering entity functioning as a token marketplace during the offering period. For example, asks and bids are not allowed to be posted on a bulletin board, and the website of the tokenized securities offering entity is not allowed to post the security token prices. However, bids are operable to be made to security token holders on the blockchain platform. For example, a bid for 5 security tokens is made, a bidding message is sent to security token holders, the security token holders review the bidding message and the bid via their digital wallets and make counter offers or direct sales (e.g., swap tokens by fulfilling contract). This process creates visibility for SEC compliance. Tokenized securities purchased during the tokenized securities offering (or through existing manual means) are tradable on a secondary market usually after a rest period.

(55) When the tokenized securities offering entity solicits to buy and sell, the entity creates a token market itself for token transfer or token exchange as the tokenizing securities offering entity will function as both a token holder and a token transactor. However, under current SEC rules at the time of the present invention, no solicitation is allowed outside allowed investors for SEC, State, and offering entity's compliance; as SEC provides for expansion, modification, or new rules, the platform of the present invention is intended to correspondingly provide for the same. In one embodiment, the tokenized securities offering entity also calculates taxes due to revenue department based on information including gross amounts of security tokens and stored information of capital gains and losses by token holders. The information is transmitted to an IRS network, which receives the information and collects taxes from security token holders.

(56) The tokenized securities offering entity keeps a log of events impacting contracts. The token transfer framework is piggybacked on logging methods used by the tokenized securities offering entity. Notifications of what is available on the token market of the tokenized securities offering entity are created based on the logging methods and sent to all token holders. The notifications are not sent outside of the whitelist of allowed investors so that the notifications are not general solicitation but rather managed by the offering entity.

(57) Books, records, contracts, and financials of the tokenized securities offering entity are accessible to invested investors for inspection. When new blockchain securities are created, the invested investors receive messages for inspection. Every invested investor has unique documents with personalized watermarks. Preferably, a web-based portal is provided, the portal having a GUI for at least one investor to manage its activity and initiate new activity within the blockchain-based platform. Contrary to a whitelist, the tokenized securities offering entity also creates and automatically maintains a blacklist that blocks domain names, individuals, competitors and others with malicious intent from the token purchase or exchange.

(58) BITE Tokens

(59) In one embodiment, tokenized securities offered by the tokenized securities offering entity are named as Blockchain Instrument for Transferrable Equity (BITE) tokens. BITE tokens can represent various forms of securities used by entities, such as common stock, preferred stock, options, warrants, convertible notes, restricted stock units and employee stock options. For example, the BITE tokens have, but is not limited to the following features:

(60) (a) Individual Optional Conversion: At any time on or after the end of a purchase period, qualified holders are enabled to elect to convert their BITE tokens into shares of voting or non-voting stock. Initially each BITE token is convertible into one share of voting or non-voting common stock. In a preferred embodiment, the conversion is subject to later proportional increase or decrease to reflect any changes to the number of outstanding shares of common stock of the offering entity attributable to any stock dividend, split, reverse split recapitalization or similar event.

(61) (b) Dividend and Distribution Equivalent Rights: qualified holders will receive payments equal to the dividends and other distributions that are paid to holders of any class of common stock on an as-converted basis.

(62) (c) Majority Approved Financing: all outstanding BITE tokens will automatically convert into a new class of securities issued by the BITE offering entity in the event of (1) the BITE offering entity offers such conversion option to the holders of BITE tokens and (2) such conversion is approved by a majority of qualified holders. The majority of qualified holders are the qualified holders who own a majority of the then outstanding BITE Tokens held by all qualified holders.

(63) (d) Liquidity Event Option: each qualified holder has the right to choose to receive other cash or non-voting common stock when a liquidity event (e.g., IPO or Change of Control) or dissolution occurs. If the liquidity event is an IPO, holders of BITE tokens will receive voting common stock instead of non-voting common stock.

(64) (e) Automatic Conversion Upon a Qualified Equity Financing: If a qualified equity financing occurs, all outstanding BITE tokens will automatically convert into the same class of securities issued in the qualified equity financing, subject to holders executing qualified equity financing documents.

(65) In one embodiment, the systems and methods are designed to comply with the regulation environment in the US. In other embodiments, the systems and methods can be used for unregulated environments or different regulation environments in other jurisdictions.

(66) FIG. 1 illustrates a system for purchasing security tokens according to one embodiment of the present invention. The system includes at least the following components: a user device for at least one investor, server computers of qualifying agencies and their supporting agencies, a server computer of the BITE token offering entity, and a blockchain platform hosting BITE token contracts in network communication, wherein the components are operatively communicative over at least one network. A web-based portal having an interactive GUI is provided for facilitating interaction via the user device and/or server computer(s).

(67) The user device for an investor sends an accreditation request message to a server computer of a qualifying agency via a GUI. The server computer of the qualifying agency issues an accreditation certificate based on information provided by the investor via the user device and information regarding the investor from supported agencies. The user device provides the accreditation certificate Identification (ID) and the qualifying agency name when registering with the server computer of the BITE token offering entity besides other required information including but not limited to legal name, address of residence, and e-mail address. In one embodiment, self-accreditation of the at least one investor is provided via the platform, based upon inputs received from the user device. The server computer of the BITE token offering entity is operable to access and synchronize with the accreditation information from different qualifying agencies via API and referral, and automatically create an up-to-date whitelist of accredited investors based on the accreditation information from different qualifying agencies. The whitelist of accredited investors includes legal names, addresses of residence, cryptocurrency accounts and/or wallet IDs, accreditation expiration dates and other essential information of the accredited investors. In one embodiment, non-accredited investors are enabled to participate under crowd-sourcing, or other alternatives as provided for based upon rules of the SEC or governing body. The server computer of the BITE token offering entity is operable to verify the accreditation certificate obtained by the investor by looking up in the whitelist of the accredited investors. In one embodiment, the whitelist is recorded and updated on the blockchain platform hosting BITE token contracts. In one embodiment, the tokenized securities offering entity also creates and maintains a blacklist and blocks domain names, individuals, competitors and others with malicious intent from the token purchase or exchange. In one embodiment, the blacklist is also recorded and updated on the blockchain platform hosting BITE token contracts. The user device for the investor is operable to send an “agree-to-terms” message if the investor intends to participate in the BITE token offering.

Then the user device receives and reviews documents related to the BITE token contract deployed on the blockchain. The user device makes the purchase of the at least one BITE token contract on the blockchain platform by transmitting a required amount of currency.

(68) In one embodiment, the terms in the BITE token offering include a maximum number of token holders, a minimum amount of investment, a number of days to rest, and a restocking fee. In one embodiment, the maximum number of token holders complies with SEC rules, for example, the maximum number of token holders is set at 1500. For example, the minimum amount of investment is an amount of cryptocurrency of \$5000 equivalent value, or other value based upon another cryptocurrency or another currency. The number of days to rest defines a time period during which BITE tokens are not allowed to be resold for inside trading restriction. In one embodiment, the number of days to rest is 60. In another embodiment, the number of days to rest is not bounded by a date, but rather until all of the tokens have been sold. The number of days to rest ensures that the security tokens are not resold after being purchased during the BITE token offering, unless allowed by the governing entity or rules body. The restocking fee is a percentage of the prices of BITE tokens being returned from an investor during a certain period of time due to various reasons with SEC compliance. For example, while the BITE token offering is open, an investor problem is identified after investment, the server computer of the BITE token offering entity is operable to reverse the BITE tokens held by the problem investor back to the BITE token offering entity, and the BITE tokens are reintroduced to the market. The server computer of the BITE token offering entity is operable to charge the problem investor for restocking the security tokens into the market. In one embodiment, BITE tokens are returned from an investor up to 5 days before the BITE token offering is closed, and a 15% restocking fee is charged.

(69) In one exemplary embodiment, the BITE token offering entity initially offers a predetermined number of tokenized securities, for example, 3000 tokenized securities, the maximum number of BITE token holders is 1000, the minimum investment of each investor is \$5000 equivalent Ethers, and the total investment is \$15,000,000 equivalent Ethers.

(70) FIG. 2 is a timeline throughout a life cycle of BITE tokens according to one embodiment of the present invention. The timeline starts with deploying BITE token contracts on a blockchain platform. There are four periods: presale period, offering period, rest period, and trading period. Dates for those periods are programmable based upon how they work together in the BITE token contracts. Variables regarding the BITE token offering are programmable on the blockchain platform. For example, the variables include start and stop dates for each period, presale prices, offering prices, minimum investment amount, and thresholds which trigger the next period. A threshold can be a date, a number of tokens purchased, or a number of investors invested.

(71) The first period is a presale period. During the presale period, potential investors are enabled to register and review documents related to the BITE token offering, and a discount is offered for purchasing BITE tokens. An initial price for tokenized securities is set by the BITE token offering entity, and the discount is based on settings of price adjustment and adjustment frequency in the BITE token contracts deployed on the blockchain platform. In one embodiment, an initial price is set as 17 Ethers per BITE token, price adjustment is set at 1 Ether, and there are 4 adjustments during a four-week presale period. In another embodiment, accredited investors receive at least one discount rate based upon an initial price with at least one time factor; by way of example and not limitation, accredited investors automatically receive 15% off of the initial price during the first week of presale, 10% off during the second week of presale, 5% off during the third week of presale, and no discount during the fourth week.

(72) The second period is an offering period. During the offering period, accredited investors are enabled to register for participation, review documents, accept terms, and purchase BITE tokens. BITE token contracts are let to invested investors and recorded on the blockchain platform. During the offering period, some investors are rejected due to failed accreditation, being a competitor of the BITE token offering entity, or being affiliated with a competitor of the BITE token offering

entity. Rejected investors are allowed to petition and demonstrate to the server computer of the BITE token offering entity that they are valid investors to the BITE token offering entity. The server computer of the BITE token offering entity is operable to verify and reinstate the rejected investors, and authorize them to participate in the BITE token offering.

(73) After the offering period, there is a rest period. The rest period is to ensure that investors participated in the BITE token offering are not brokers or dealers as defined by the SEC rule 144, or other rules or regulations by the SEC or other governing authority.

(74) After the rest period, it is a trading period, during which BITE tokens are tradable on a secondary market. This period is able to be bounded by a time requirement. The server computer of the BITE token offering entity is still operable to enforce SEC rules for compliance on the blockchain platform. For example, the server computer of the BITE token offering entity is operable to suspend certain investors or trading activities for a certain time period.

(75) In one embodiment, the BITE token offering entity is operable to trade BITE tokens as an exchange automatically based on terms in the BITE token contracts. In one embodiment, asks and bids are performed in order. There is only one bid or one ask per allowed investor at any given time such that they cannot sell tokenized securities that they no longer own, or attempt to purchase tokenized securities when they do not have available funds.

(76) Once the BITE token contract is let, it is not changeable. However, terms within the contract that are executable are changeable, for example, parameters within the contract (e.g., suspension of trading or investor).

(77) FIG. 3 illustrates interactions between different entities during a security token offering according to one embodiment of the present invention. A server computer of the security token offering entity deploys security token contracts and raise Ethers on ETHEREUM blockchain. The server computer of the security token offering entity sends a request message to a server computer of crowdsourcing investor bulletin (IB) indicating that the token offering entity wants investment and access to investor information. The server computer of the crowdsourcing IB sends a reply message allowing or disallowing the token offering entity to access to the investor information. A user device for an investor who intends to participate in the security token offering requests qualification from the crowdsourcing IB via a GUI. If the requesting investor is qualified as an approved (for example, if the investor's income must be verified) or accredited investor, the server computer of the crowdsourcing IB sends a qualification certificate to the user device for the investor. A blockchain oracle is utilized by the server computer of the token offering entity to look up investor qualification information from the crowdsourcing IB to verify investor accreditation status. Once the investor accreditation status is verified, the user device is operable to review and accept the security token contract, and send Ethers from an ETHEREUM account to finish purchasing security tokens on the ETHEREUM blockchain.

(78) FIG. 4 illustrates a token purchasing process on blockchain during a security token offering according to one embodiment of the present invention. A user device for an investor communicates with a server computer of an accreditation service to get the investor approved as an accredited investor to participate in the security token offering. The user device includes an ETHEREUM wallet of the investor. The ETHEREUM wallet includes a unique identifier from which Ethers are sent for token purchasing on the ETHEREUM blockchain. The user device transmits investor information to the server computer of the security token offering website, with the investor information including an email address, a uniquely identified account, an encrypted code for accreditation status, and a third-party verifying agency who issued the encrypted code for accreditation status to the server computer of the security token offering website. The server computer of the security token offering website registers the accredited investor on the ETHEREUM blockchain with the information received from the user device. The user device receives documents included in a token contract after registering the investor with the sever computer of the security token offering website. For example, but not for limitation, the documents

include PPM, subscription agreement, Intellectual Property, and Article(s) of Incorporation. The user device transmits an acceptance message after reviewing documents included in a security token contract and sends Ethers to purchase security token contract on the blockchain.

(79) FIG. 5 is a timeline throughout a life cycle of tokenized securities according to one embodiment of the present invention, wherein the life cycle includes at least one time interval. By way of illustration, there is a promotion period before the tokenized securities offering. The platform of the present invention is operable to receive applications by at least one intended investor for accreditation, in order to participate in the tokenized securities offering during the promotion period. During the offering period, the platform is operable to provide for the purchase of tokens on a blockchain platform by accredited investors. In one exemplary embodiment of a time interval, the offering period of the tokens lasts 30 days, with a rest period following the offering period that lasts 90 days.

(80) A verification process is required in an electronic system for a tokenized securities offering to demonstrate that a potential investor is an accredited investor according to SEC rules. After a rest period, existing shareholders are enabled to sell to new investors who are not required to demonstrate an accredited investor status. In one embodiment, the systems and methods of the present invention allow self-verification by new investors (e.g., by checking a box "I'm an accredited investor based on income or assets . . ." and provide qualification documents), and the tokenized securities offering entity performs qualification assessment (QA) manually or automatically during the offering period and rest period instead of using a third-party service for verification. After the rest period, there is a resale period when tokens are traded between existing tokenized securities holders and new investors or other existing holders.

(81) FIG. 6 shows an exemplary illustration for transactions and contracts recorded in blocks on the ETHEREUM blockchain, although the present invention is operable for at least one blockchain, including other blockchain platforms. Continuing with the example of FIG. 6, the ETHEREUM blockchain, every block includes a multiplicity of transactions and a hash linked to a preceding block. Each transaction includes an account identifier, an ether balance, a transaction date, and data. In one embodiment, the data in each transaction includes security token contracts deployed by the systems and methods of the present invention. Each tokenized securities contract includes a contract ID, an Ether account, a balance of tokens in an escrow account for the tokenized securities offering entity on the blockchain, and a state of the token holder. The state of the token holder is based on the whitelist when the token contract is let. In one embodiment, each tokenized securities contract also include data and manipulation methods.

(82) FIG. 7 illustrates transactions for a token contract between an investor and a security token offering entity on the ETHEREUM blockchain. In one embodiment, a token contract is 20 Ethers at a time during a token offering. When token contracts offered by the token offering entity are deployed on the ETHEREUM blockchain, there are 3000 contracts available, and one of the at least one investor is identified as cd136248562, which has 300 Ethers in his digital wallet, which are both recorded in one block on the ETHEREUM blockchain. Investor cd136248562 registers with the token offering entity for token purchase, one token contract is let to the investor, and the transaction is recorded in one block on the ETHEREUM blockchain. The transaction includes a balance of investor cd136248562 as 280 Ethers, and the token contract. The token contract includes the balance of tokens in an escrow account for the token offering entity on the ETHEREUM blockchain.

(83) FIG. 8 lists investor documents in a token purchase process according to one embodiment of the present invention. When a user device registers an investor on the server of the tokenized securities offering entity website, the user device needs to send an acceptance message regarding a non-disclosure agreement and provide an e-mail address of the investor to the server of the tokenized securities offering entity. In order to proceed in the purchase process, the user device needs to provide information including a unique qualification ID for investor accreditation and its

issuing agency to the server of the tokenized securities offering entity website. Meanwhile, the server of the tokenized securities offering agency provides investor documents available for download on a blockchain platform. After downloading the investor documents, the user device of the investor sends an agreeing message regarding the terms and conditions specified in an PPM and a subscription agreement, and provides a unique identifier address to make the purchase. Investor documents include PPM, subscription agreement, and inventor qualification documents. The PPM specifies risks, company charter, business plan, etc. The PPM is not required for investor signature. The subscription agreement includes introduction of the tokenized securities contracts offered by the tokenized securities offering entity and stock definition. The subscription agreement is required for investor signature. The investor qualification documents provide proof of the investor's net income or annual income with investor signature for self-verification. In one embodiment, the server computer of the tokenized securities offering entity sends a message asking for an accreditation status of new investors participating in the tokenized securities offering. The new investors are enabled for self-verification by submitting qualification documents instead going through a third-party accreditation agency. In one embodiment, existing shareholders are required to confirm their accredited investor status on an annual basis. The server computer of the tokenized securities offering entity sends a message asking for renewed accreditation status of the existing shareholders. The existing shareholders are enabled for self-verification by submitting qualification documents instead going through a third-party accreditation agency

(84) FIG. 9 illustrates a tokenized securities offering timeline according to one embodiment of the present invention, showing exemplary timelines of promotion through offering and rest periods. Alternative timelines or intervals are provided in other embodiments. Social media outreach, press releases, and other marketing tools are utilized to raise awareness for the tokenized securities offering on the platform, and it lasts 4 weeks before a promotion starts. The promotion lasts 2 weeks, during which the tokenized securities offering website is live for promotion. Then token contracts are deployed on a blockchain platform and the price is set for the token contract. There is a 2-week presale period before a 2-week official tokenized securities offering. After the official offering, there is a rest period before the token contracts can be traded. In this embodiment, the rest period is 1 year. After a rest period, trading begins for buying and selling the token contracts. In one embodiment, the present invention provides an algorithm for price setting. Various variables are programmed in the token contracts on the blockchain platform. For example, the price is set at lowest at the beginning of the offering, but a minimum amount of purchase is required. Because of the volatility of cryptocurrencies and other currencies, the present invention provides an automated mechanism to dynamically set the price of token contracts based on the cryptocurrency price, by way of example and not limitation, the Ether price, when the contract is about to let. The token contract price is locked in at the time of purchasing and not changeable afterwards even though Ether price changes.

(85) FIG. 10 is a flowchart of a token purchase process according to one embodiment of the present invention. A server computer of a tokenized securities offering entity provides a tokenized securities offering landing page, which includes promotion materials, offer status, and a work flow of token purchase. A user device for an investor transmits required investor information to the server computer of the tokenized securities offering entity and accepts terms of the offering website (e.g., non-disclosure and privacy). In one embodiment, the registration process includes 2-step authentication or other authentication protocols to effect security. In another embodiment, the registration is enabled with Open ID protocol. The user device is enabled to use a third-party login of the investor, and the third-party login preferably has 2-step authentication as well.

(86) If the user device provides a qualification ID of the investor's accreditation status to the server computer of the tokenized securities offering agency, the server computer of the tokenized securities offering agency is operable to verify the investor's accreditation status by comparing information related to the investor to a whitelist and a blacklist the server computer of the offering

entity maintains up to date. If the investor does not have a qualification ID when registering, the user device is redirected to a third-party website for accreditation, or self-accreditation. If the user obtains a qualification ID from the third-party website for accreditation, the offering website is able to verify his or her accreditation status based on the up to date whitelist and blacklist maintained by the offering. The whitelist includes information related to accredited investors, wherein the information includes but is not limited to investor names, addresses, at least one qualification ID and corresponding issuing agency. The blacklist includes affiliations, competitors and US addresses, other addresses the offering entity wants to block, predetermined foreign investors, and/or bad actors based on data obtained from OFAC and/or other databases.

(87) Once the server computer of the offering entity verifies the investor as an accredited investor, the user device of the investor receives a link to documents included in a token contract for reviewing and accepting. The user device transmits a Wallet ID which is a uniquely identified account and a full signature and accepts token offering terms, SEC rules, and non-disclosure agreements (NDA) after reviewing the documents. The user device then makes transactions for purchasing the token contract by transmitting a certain amount of cryptocurrency from the investor's digital wallet to an escrow account for the token offering entity on a blockchain platform.

(88) The process of purchasing tokenized securities contracts is executed automatically based on the contract deployment on the blockchain platform. In one embodiment, manual verification is needed for blocking competitors and other rule-complying purposes.

(89) In one embodiment, the present invention provides systems and methods for implementing tokenized securities on a blockchain based on ERC20 token standard (or similar subsequent standard). The present invention requires at least one uniquely identified account on the blockchain platform for an investor to participate in the tokenized securities offering. Cryptocurrencies accepted by the offering entity are selected from the group consisting of Bitcoin, ETHEREUM, LITECOIN, and other cryptocurrencies of liquid value. Credit card, debit card, cash, or other traditional payment methods are not accepted during a cryptocurrency only offering. In one embodiment, the offering entity computer is in network communication with certain cryptocurrency exchanges or traditional banking system to facilitate token purchases via fiat currencies. In one embodiment, a centralized wallet and trustee is utilized as a special purpose vehicle to facilitate investors who do not have accepted cryptocurrencies, or uniquely identified accounts on the blockchain-based security token platform required to purchase tokenized securities during an offering as a single person entity. Investors who do not have accepted cryptocurrencies or uniquely identified accounts pay the centralized wallet and trustee by fiat currencies or other payment methods accepted by the centralized wallet and trustee via an off-chain transaction.

(90) In one embodiment, security tokens are tokenized securities. In one embodiment, security tokens represent tokenized securities. In one embodiment of the present invention, tokenized securities contracts or security token contracts represent securities. In one embodiment, the tokenized securities contracts are transformed into other formats during their life cycles after the tokenized securities offering is closed. In another embodiment, tokenized securities contracts can have children contracts. Children contracts are those that can manipulate or otherwise manage the tokenized securities created by the parent contract. An initial tokenized securities offering entity is operable to launch secondary tokenized securities offerings.

(91) Contracts

(92) A standard smart contract in any blockchain contains information including a contract owner, a contract address, contract data, and terms (e.g., rules and methods to act on the contract).

(93) With ordinary blockchain smart contracts, the contract owner remains the same for the entirety of the contract lifecycle (i.e., from deployment until termination conditions). In contrast, the tokenized securities contracts in the present invention are transferrable. Every contract has a unique identifier. In one embodiment, the unique identifier is a hash value generated by a hash algorithm in the blockchain. In one embodiment, the unique identifier is generated by other mathematical

algorithm (e.g., mathematical formulation of quantum mechanics).

(94) In one embodiment, the blockchain-based security token platform of the present invention provides a proxy contract between party A and party B. FIG. 11 illustrates a proxy contract according to one embodiment of the present invention. Party A participates on the tokenized securities offering blockchain directly as the proxy owner. In one embodiment, party A is the platform owner who offers and/or trades tokenized securities as a proxy of party B. In one embodiment, party A is a broker/dealer offering and/or trading tokenized securities on the blockchain-based security token platform. Party B can be any company or entity intending to offer and/or trade tokenized securities on the blockchain-based security token platform via a proxy. A proxy contract includes a unique contract address, a unique address of the proxy owner, a unique address of the company X, contract data which is being proxied, and terms to act on the proxy contract. Terms of the proxy contract include a time the proxy contract is signed, renewals, terminations, and methods to act on the contract data. The methods to act on the contract data includes but not limited to changing ownership from the contract originating party to another party (e.g., company X) by changing the proxy owner unique identifier to company X. With the proxy contract, party A can represent party B for participating on the blockchain-based security token platform. The proxy contract allows for ownership to be exchanged on specific contract terms being met or contract completion (all the terms are met). The proxy contract provides the means to pass ownership of a contract from one party to another. In another embodiment, proxy contracts are used to form a multi-party transaction where the ownership of the contract is transferred to another party that represents a group of new owners that have been aggregated.

(95) With ordinary blockchain smart contracts, the data in the contract is visible to every node in the network, but only the contract owner can manipulate the data. In one embodiment, the blockchain-based security token platform of the present invention provides an escrow contract between party A (proxy owner) and party B (company X). An escrow contract encrypts the 'important data' using the owner's unique identifier address so that only the owner of the contract can use its private key and/or password to view the 'important' data. An escrow contract typically inherits the functionality of a proxy contract to enable the ownership of the escrow contract to change when terms of the escrow agreement between the parties have been met. An escrow contract holds onto valuable information (e.g., a digital key to a lockbox, account identifier, disclosure information). The owner of the escrow contract has access to the escrowed information. In one embodiment, a proxy contract is used in conjunction with an escrow contract (or other types of contracts identified below) to allow the escrowed information to be transferred from one party to another. For example, a proxy contract is made between two parties that are settling a BITE transaction outside of the blockchain, and the escrow contract is holding the bank account information from the buyer and is owned by the seller. Once the proxy contract changes ownership (e.g., because of release of information by the bank that funds are available as captured by a software oracle), the data is decrypted by the original owner and re-encrypted using the new owner's uniquely identified account which make the escrowed information visible to the new owner (buyer) when they use their key and/or password to access the escrowed information. FIG. 12 illustrates an escrow contract according to one embodiment of the present invention. The escrow contract ensures data security. In one embodiment, the escrow contract is used for off-chain transactions. For example, automated clearing house (ACH) transaction data can be held at the escrow contract with encryption and accessible only by an owner's decryption key and/or password.

(96) In one embodiment, the blockchain-based security token platform of the present invention provides an aggregate contract. FIG. 13 illustrates an aggregate contract according to one embodiment of the present invention. The aggregate contract is between a party A and a multiplicity of parties including investors and parties included in proxy contracts, escrow contracts, and any other contracts. The aggregate contract creates a group to act as a single entity represented

by party A. The aggregate contract includes a list of uniquely identified accounts pointing to different individuals, aggregate contracts, or other contracts, and allows the multiplicity of parties and party A to have transactions in addition to the transactions that party A participates on behalf of the multiplicity of parties. For example, a group of tokenized security holders could form an aggregate contract whereby Party A is a legal entity (such as an LLC), that can be used to sell the entirety of the group's tokenized securities for a single price to another party in a proxy contract. Once that transaction is complete, the aggregate contract between the multiplicity of parties and party A automatically exchanges the funds in the correct amounts from party A to each member of the group based on the tokenized securities they contributed to the aggregation, less any transaction fees, etc.

(97) In one embodiment, an aggregate contract is operable to represent a multiplicity of levels of contracts, including but not limited to nested levels of aggregate contracts. An aggregate contract is operable to hold or retain a unique identifier to another aggregate contract, which in turn retains a unique identifier to another aggregate contract, and so on. An aggregate contract is further operable to represent an investor, who is actually an entity (i.e., private equity fund), which has many investors, of which one of those investors could represent another entity (i.e., trust fund), which is made up of several parties.

(98) In one embodiment, a typical contract structure on the blockchain-based security token platform of the present invention is in the format of a proxy contract representing tradable securities. FIG. 14 illustrates a typical contract according to one embodiment of the present invention. In one embodiment, the typical contract is a tradable security (i.e., BITE) which inherits transfer ownership capabilities from a proxy contract. Tradable securities contain unique identifiers representing different contracts and libraries, including aggregate contracts, proxy contracts, and etc. In one embodiment, a tradeable security represents a common stock, which has an aggregate of investors. In one embodiment, some of the investors are nested levels of aggregates of investors (i.e., fund, legal entity), and other investors are represented by dealers or brokers in a proxy contract. Additionally, there is one or more libraries assisting the performance of data type checks, data storage, provide notification of events happening within the contracts, etc. The tradeable security has rules which govern allowed parties in the sale (e.g., whitelist and blacklist), timing of sales (e.g., rest periods), disclosure restrictions (i.e., who can view the documents and whether they are mandatory or not), allowed price boundaries (e.g., holding company allows sales over a predetermined price), transaction fees (e.g., platform, broker, dealer, discounts), and contract controls (e.g., start, suspend, resume, terminate).

(99) In one embodiment, the blockchain-based security token platform of the present invention is operable to link contracts with the same owner by using one single proxy address, which can access to contract data and terms in the linked contracts. Linking contracts in this way allows deployment of different contracts at different times to work together on a single transaction. This capability essentially allows for rules to be added to a transaction after initial deployment without modification to the original contract.

(100) In one embodiment, the blockchain-based security token platform of the present invention is also operable to suspend a contract. Suspending a contract disables the ability for a contract to be executed. In one embodiment, it enables an offering entity to halt the purchase of new tokenized securities because there has been a significant event or material change to the offering entity. In another embodiment, it enables a holding entity to suspend a proxy contract because the holding entity finds out that a broker or dealer is representing a competitor. Suspended contracts can be resumed (allowed to continue execution) or terminated (no longer able to execute). In yet another embodiment, it enables the SEC to halt a tokenized securities trading under investigation.

(101) In one embodiment, the blockchain-based security token platform of the present invention also provides a voting contract between an entity and its shareholders/investors for proxy voting. In one embodiment, the voting contract inherits an aggregate contract. The voting contract includes a

list of questions for shareholders/investors. The entity deploys voting contracts on the blockchain-based security token platform. Each investor who holds at least one tokenized security from the entity receives a voting contract with a unique address. Once the investor submits the answers to the list of questions in the voting contract, the voting contract is collected, processed, counted, and recorded on the blockchain. In one embodiment, the shareholders are aggregated, and the present invention provides for each of the nested votes to be counted with rules applied to determine whether a simple majority, super majority, or some other calculation is required to provide the vote for the represented party. In cases of proxy contracts, the rules in the proxy contracts govern whether the current owner of the proxy contract votes or whether the proxied party votes.

(102) The blockchain-based security token platform of the present invention allows for peer-to-peer trading of tokenized securities, enables private equities transferrable, increases liquidity of private equities, automates regulatory compliance, enforces an entity's desire for limited disclosure, and reduces risk of non-compliance. In addition, proxy voting and self-accreditation enabled by the blockchain-based security token platform also facilitates engagement of peer shareholders among different issues.

(103) In one embodiment, all tokenized securities contracts fulfill all the rules constructed by or established by the SEC for registration exemptions, for example, including but not limited to Section 4(a)(2), Regulation D, Regulation Crowdfunding, Regulation A, Intrastate, and any subsequent rules created by the SEC.

(104) In one embodiment, the tokenized securities contracts fulfill all state rules for compliance. For example, under Regulation D rule 506(c), a company raising private money preempts a state registration, but for Regulation D rule 504, the contract would also have to ensure the company has registered with the state.

(105) In one embodiment, the tokenized securities contracts fulfil any arbitrary company rule. For example, it is required that a representative of a company approves an equity sale before payment and ownership of a tokenized securities contract can be exchanged.

(106) In one embodiment, the tokenized securities contracts are grouped by certain rules applied to the tokenized securities contracts. For example, a company has several classes of securities (e.g., Series A, Series B, etc.) in the company's entire list of equities. A rule for a private company is that the maximum number of shareholders for all securities must be less than 2000. When offering tokenized securities contracts, the company groups the tokenized securities contracts based on the rule mentioned above.

(107) In one embodiment, the tokenized securities contracts are linked by common ownership hashes and keys, which are operable to manage the contracts through their entire lifecycle including deployment, usage, and retirement. Thus, tokenized securities contracts deployed at different times are operable and can participate together in one transaction.

(108) In one embodiment, the tokenized securities payments are made outside of the blockchain-based security token platform of the present invention. For example, an ACH payment can be executed outside of the blockchain-based security token platform and verified by the tokenized securities contract, and then the tokenized securities are transferred.

(109) In one embodiment, the blockchain-based security token platform of the present invention comprises a blockchain recording security token transactions from the first block (the genesis block) until the current block. In one embodiment, the blockchain is designed to retire as of a certain block such that nodes are synchronized without holding all the data from the retired part. In other words, a blockchain can be retired at a prescribed interval (e.g., block No. 10,000) and any new nodes added to the blockchain can start at the next node (e.g., block No. 10,001 with block No. 10,000 as the genesis block). Since a blockchain is immutable, there is no need to keep previous blocks active and synchronized on all nodes. This solves the problem of data growth on all the nodes.

(110) In one embodiment, a company solicits investor, owners, and/or family members to approve

certain actions of the executive team. Solicitations and responses thereto are recorded on the blockchain-based security token platform of the present invention to provide specificity to board meeting minutes and other communications.

(111) In one embodiment, the blockchain-based security token platform is operable to create tokenized securities representing securities to previously offered to pre-existing shareholders. All the capabilities described for an initial offer apply, including, but not limited to, whitelist, blacklist, register, accepting terms, purchase agreement, purchasing, reselling. In one embodiment, BITE tokens can be offered and/or assigned to pre-existing shareholders and/or new shareholders.

(112) In one embodiment, the present invention enables existing shareholders/proxies to update their accreditation statuses on a periodic basis. For example, SEC Act of 1934, Section 12(g) requires a company with more than \$10 million in assets desiring to remain private is limited to 2,000 investors, of which only 500 can be non-accredited. Thus, for a company to remain private it must track how many of its investors are accredited on an annual basis.

(113) In one embodiment, the present invention provides federated purchases of tokenized securities. For example, a BITE token represents securities from two companies, BITE owners are enabled to exchange ownership in a like-kind exchange (e.g., no payment is made).

(114) In one embodiment, the present invention provides aggregated purchases of tokenized securities on the blockchain-based security token platform. For example, a purchaser of BITE tokens is enabled to purchase BITE tokens from two different owners with a single payment. For example, owner A has 200 BITE tokens for \$550, and owner B has 300 BITE tokens for \$800, a purchaser can buy both sets of BITE tokens in a single transaction for \$1,350.

(115) In one embodiment, the present invention provides aggregated sales of tokenized securities on the blockchain-based security token platform. For example, two sellers are enabled to offer their BITE tokens together with the same unit price.

(116) FIG. 15 is a diagram illustrating a signup process for offering tokenized securities on a blockchain platform according to one embodiment of the present invention. A tokenized securities offering entity (e.g., a broker or a company) registers on the blockchain-based security token platform. The blockchain-based security token platform then creates a proxy contract with the broker or company for a tokenized securities offering. The tokenized securities offering entity then creates tokenized securities on the blockchain-based security token platform, which deploys tokenized securities contracts for the tokenized securities offering entity. The tokenized securities offering entity then adds its investors and/or owners to the blockchain-based security token platform, which requests a capitalization table for verification from the investors and/or owners. Interested investors can register on the blockchain-based security token platform for participating in the tokenized securities offering. The blockchain-based security token platform then requests an up-to-date accreditation status from the registered investors. The registered investors respond by answering questions that demonstrate their accreditation status. In one embodiment, the blockchain-based security token platform creates a whitelist of accredited investors based on data extracted from multiple investor accreditation related databases.

(117) FIG. 16 is a diagram illustrating a trading process for a tokenized securities seller on a blockchain platform according to one embodiment of the present invention. The tokenized securities seller (e.g., a broker, an investor) makes a request on the blockchain-based security token platform to sell tokenized securities. The blockchain-based security token platform then checks approval from the company who offered/issued the tokenized securities, checks rules for trading tokenized securities for compliance, and authorizes the seller to sell. The tokenized securities seller then posts an ask on the blockchain-based security token platform, a buyer (e.g., a broker, an investor) accepts the ask on the blockchain-based security token platform. If a buyer doesn't agree with the ask price, they can post a bid price back to the seller. The seller and buyer can send bid and ask prices back and forth until there is a mutual agreement where one party accepts the price of the other party. Once agreement has been made with the acceptance of an ask or bid price, the

blockchain-based security token platform then checks rules and creates an escrow contract with the seller and the buyer, performing as an escrow account for the seller and the buyer. The buyer makes payment to the escrow account, and the seller transfers the tokenized securities the buyer intends to buy to the escrow account. The escrow account checks rules and processes the payment, and then release the payment to the seller and the tokenized securities to the buyer.

(118) In one embodiment of the present invention, the blockchain-based security token platform is operable to modify a lifecycle of a contract, including but not limited to start, suspend, resume, or terminate. In one embodiment, the proxy contract enables the contract ownership transfer from one unique identifier to another unique identifier based on individual contractual terms being met or the contract completion. In one embodiment, an escrow contract securely hides information from participating parties other than the owner. In one embodiment, during the trading or transferring stage, the server computer of the tokenized securities offering entity is operable to set minimum and/or maximum prices, timing, and approval on sales (or resales) of the tokenized securities. In one embodiment, the server computer of tokenized securities offering entity is operable to manage access control for the information related to the offering, trading, and transferring, for example but not for limitation, if a potential tokenized securities purchaser can view or have access to disclosure documents including, but not limited to, accreditation, brokers, dealers, employees, corporate officers, competitors, location country, location state, income level.

(119) Referring now to FIG. 17, a schematic diagram illustrating a cloud-based computing network used in one embodiment of the invention for interactions between user devices and a server computer of a tokenized securities offering entity is shown. As illustrated, components of the systems and methods include the following components and sub-components, all constructed and configured for network-based communication, and further including data processing and storage. As illustrated in FIG. 17, a basic schematic of some of the key components of a financial settlement system according to the present invention are shown. The system **200** comprises a server **210** with a processing unit **211**. The server **210** is constructed, configured and coupled to enable communication over a network **250**. The server provides for user interconnection with the server over the network using a personal computer (PC) **240** positioned remotely from the server, the personal computer having instructions **247**. Furthermore, the system is operable for use with at least one or a multiplicity of remote computers, computing devices, or terminals **260, 270**, having operating systems **269, 279** or software operable thereon. For example, a client/server architecture is shown. Alternatively, a user is able to interconnect through the network **250** using a user device such as a personal digital assistant (PDA), mobile communication device, or mobile computing device, such as by way of example and not limitation, a mobile phone, a cell phone, smart phone, tablet computer, laptop computer, wearable computing device, netbook, a terminal, or any other computing device suitable for network communication, whether wired or wireless. Also, alternative architectures are able to be used instead of the client/server architecture. For example, a PC network, or other suitable architecture are able to be used. The network **250** includes the Internet, an intranet, or any other network suitable for searching, obtaining, and/or using information and/or communications. The system of the present invention further includes an operating system **212** installed and running on the server **210**, enabling server **210** to communicate through network **250** with the remote, distributed user devices. The operating system is any operating system known in the art that is suitable for network communication as described hereinbelow. Data storage **220** houses an operating system **222**, memory **224**, and programs **226**.

(120) Additionally, or alternatively to FIG. 17, FIG. 18 is a schematic diagram of an embodiment of the invention illustrating a computer system and network, generally described as **800**, having a network **810** and a plurality of computing devices **820, 830, 840**. In one embodiment of the invention, the computer system **800** includes a cloud-based network **810** for distributed communication via the network's wireless communication antenna **812** and processing by a plurality of mobile communication computing devices **830**. In another embodiment of the

invention, the computer system **800** is a virtualized or cloud-based computing system capable of executing any or all aspects of software and/or application components presented herein on the computing devices **820, 830, 840**. In certain aspects, the computer system **800** is implemented using hardware or a combination of software and hardware, either in a dedicated computing device, or integrated into another entity, or distributed across multiple entities or computing devices.

(121) By way of example, and not limitation, the computing devices **820, 830, 840** are intended to represent various forms of digital computers **820, 840, 850** and mobile devices **830**, such as a server, blade server, mainframe, mobile phone, a personal digital assistant (PDA), a smart phone, a desktop computer, a netbook computer, a tablet computer, a workstation, a laptop, and other similar computing devices. The components shown here, their connections and relationships, and their functions, are meant to be exemplary only, and are not meant to limit implementations of the invention described and/or claimed in this document.

(122) In one embodiment, the computing device **820** includes components such as a processor **860**, a system memory **862** having a random access memory (RAM) **864** and a read-only memory (ROM) **866**, and a system bus **868** that couples the memory **862** to the processor **860**. In another embodiment, the computing device **830** additionally includes components such as a storage device **890** for storing the operating system **892** and one or more application programs **894**, a network interface unit **896**, and/or an input/output controller **898**. Each of the components is coupled to each other through at least one bus **868**. The input/output controller **898** receives and processes input from, or provide output to, a number of other devices **899**, including, but not limited to, alphanumeric input devices, mice, electronic styluses, display units, touch screens, signal generation devices (e.g., speakers) or printers.

(123) By way of example, and not limitation, the processor **860** is a general-purpose microprocessor (e.g., a central processing unit (CPU)), a graphics processing unit (GPU), a microcontroller, a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA), a Programmable Logic Device (PLD), a controller, a state machine, gated or transistor logic, discrete hardware components, or any other suitable entity or combinations thereof that can perform calculations, process instructions for execution, and/or other manipulations of information.

(124) In another implementation, shown in FIG. **18**, a computing device **840** uses multiple processors **860** and/or multiple buses **868**, as appropriate, along with multiple memories **862** of multiple types (e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core). Also, multiple computing devices are connected via at least one network, with each device providing portions of the necessary operations (e.g., a server bank, a group of blade servers, or a multi-processor system).

Alternatively, some steps or methods are performed by circuitry that is specific to a given function.

(125) According to various embodiments illustrated in FIG. **18**, the computer system **800** operates in a networked environment using logical connections to local and/or remote computing devices **820, 830, 840, 850** through a network **810**. A computing device **830** connects to a network **810** through a network interface unit **896** connected to the bus **868**. Computing devices communicate communication media through wired networks, direct-wired connections or wirelessly such as acoustic, RF or infrared through a wireless communication antenna **897** in communication with the network's wireless communication antenna **812** and the network interface unit **896**, which include digital signal processing circuitry when necessary. The network interface unit **896** provides for communications under various modes or protocols.

(126) In one or more exemplary aspects, the instructions are implemented in hardware, software, firmware, or any combinations thereof. A computer readable medium provides volatile or non-volatile storage for one or more sets of instructions, such as operating systems, data structures, program modules, applications or other data embodying any one or more of the methodologies or functions described herein. The computer readable medium illustrated in FIG. **18** includes the

memory **862**, the processor **860**, and/or the storage media **890** and is a single medium or multiple media (e.g., a centralized or distributed computer system) that store the one or more sets of instructions **900**. Non-transitory computer readable media includes all computer readable media, with the sole exception being a transitory, propagating signal per se. The instructions **900** further are transmitted or received over the network **810** via the network interface unit **896** as communication media, which includes a modulated data signal such as a carrier wave or other transport mechanism and includes any delivery media. The term “modulated data signal” means a signal that has one or more of its characteristics changed or set in a manner as to encode information in the signal.

(127) Storage devices **890** and memory **862** illustrated in FIG. **18** include, but are not limited to, volatile and non-volatile media such as cache, RAM, ROM, EPROM, EEPROM, FLASH memory or other solid state memory technology, disks or discs (e.g., digital versatile disks (DVD), HD-DVD, BLU-RAY, compact disc (CD), CD-ROM, floppy disc) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium that can be used to store the computer readable instructions and which can be accessed by the computer system **800**.

(128) It is also contemplated that the computer system **800** does not include all of the components shown in FIG. **18**, or include others components that are not explicitly shown in FIG. **18**, or utilizes an architecture completely different than that shown in FIG. **18**. The various illustrative logical blocks, modules, elements, circuits, and algorithms described in connection with the embodiments disclosed herein are implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate this interchangeability of hardware and software, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans are able to implement the described functionality in varying ways for each particular application (e.g., arranged in a different order or partitioned in a different way), but such implementation decisions should not be interpreted as causing a departure from the scope of the present invention.

(129) One or more communications protocols and/or methods for wired or wireless communications over the at least one network are able to be used with the present invention systems and methods.

(130) The network-based communication can be wired or wireless using protocols such as, by way of example and not limitation, internet protocol (IP) including IPv4 and IPv6, cellular protocols 1G, 2G, 3G, 4G/LTE, and 5G, 802.11, ZIGBEE, BLUETOOTH, or others currently available or developed in the future. Also, by way of definition and description supporting the claimed subject matter, preferably, the present invention includes communication methodologies for messaging via a communication layer or for data transmission or communication over at least one network as described in the foregoing and in the following. IP-based communications over a network are most preferred for secure transmission, and for transmission of data having at least one of a security, a priority, a transport route, and content. Correspondingly, and consistent with the communication methodologies for transmitting or communicating data from the platform or at least one server, or within a closed system, as described hereinabove, according to the present invention, as used throughout this specification, figures and claims, the term “ZIGBEE” refers to any wireless communication protocol adopted by the Institute of Electronics & Electrical Engineers (IEEE) according to standard 802.15.4 or any successor standard(s), the term “WI-FI” refers to any communication protocol adopted by the IEEE under standard 802.11 or any successor standard(s), the term “WIMAX” refers to any communication protocol adopted by the IEEE under standard 802.16 or any successor standard(s), and the term “BLUETOOTH” refers to any short-range communication protocol implementing IEEE standard 802.15.1 or any successor standard(s).

Additionally or alternatively to WIMAX, other communications protocols are able to be used, including but not limited to a “1G” wireless protocol such as analog wireless transmission, first generation standards based (IEEE, ITU or other recognized world communications standard), a “2G” standards based protocol such as “EDGE” or “CDMA 2000” also known as “1×RTT”, a 3G based standard such as “High Speed Packet Access (HSPA) or Evolution for Data Only (EVDO), any accepted 4G standard such as IEEE, ITU standards that include WIMAX, Long Term Evolution “LTE” and its derivative standards, any Ethernet solution wireless or wired, or any proprietary wireless or power line carrier standards that communicate to a client device or any controllable device that sends and receives an IP-based message. The term “High Speed Packet Data Access (HSPA)” refers to any communication protocol adopted by the International Telecommunication Union (ITU) or another mobile telecommunications standards body referring to the evolution of the Global System for Mobile Communications (GSM) standard beyond its third generation Universal Mobile Telecommunications System (UMTS) protocols. The term “Long Term Evolution (LTE)” refers to any communication protocol adopted by the ITU or another mobile telecommunications standards body referring to the evolution of GSM-based networks to voice, video and data standards anticipated to be replacement protocols for HSPA. The term “Code Division Multiple Access (CDMA) Evolution Data-Optimized (EVDO) Revision A (CDMA EVDO Rev. A)” refers to the communication protocol adopted by the ITU under standard number TIA-856 Rev. A.

(131) Certain modifications and improvements will occur to those skilled in the art upon a reading of the foregoing description. By way of example and not limitation, the description describes a blockchain-based platform for investment activity that is automatically managed by electronic smart contracts for at least one accredited investor. However, the SEC or other governing or regulatory authority, is able to provide for non-accredited participation in at least one investment opportunity, for which the platform would similarly function. In another example, while the description is focused on cryptocurrency illustrations, other currency equivalents are provided for by the present invention. The above-mentioned examples are provided to serve the purpose of clarifying the aspects of the invention and it will be apparent to one skilled in the art that they do not serve to limit the scope of the invention. All modifications and improvements have been deleted herein for the sake of conciseness and readability but are properly within the scope of the present invention.

Claims

1. A system for performing transactions of tokenized securities during a tokenized securities offering, comprising: a server computer for a tokenized securities offering entity in network communication with a securities token platform; wherein the securities token platform is configured to record tokenized securities contracts for the transactions of the tokenized securities; wherein the tokenized securities contracts comprise at least one proxy contract configured for contract ownership transfer from a first unique identifier to a second unique identifier, wherein the at least one proxy contract is recorded on an immutable ledger; wherein a first party represents a second party for participating on the securities token platform via the at least one proxy contract; wherein the securities token platform is configured to automatically maintain a whitelist of accredited investors, wherein the whitelist is recorded on the immutable ledger; wherein the securities token platform is configured to use a blockchain oracle to confirm accreditation credentials of the first party, wherein the blockchain oracle is in real-time network communication with a plurality of data sources; wherein, upon confirmation that the first party is allowed to represent the second party, the securities token platform is configured to automatically add the first party to the whitelist; wherein the server computer is operable to confirm accreditation status of the first party and the second party based on the whitelist; wherein the tokenized securities contracts

comprise encrypted information related to an owner of the tokenized securities contracts, wherein the information is encrypted using a first unique identifier associated with the owner; wherein upon a change of ownership of the at least one proxy contract associated with the tokenized securities contracts, the encrypted information is decrypted and re-encrypted using a second unique identifier associated with a new owner; wherein the tokenized securities contracts include a voting contract, wherein the voting contract is between each holder of the tokenized securities and a corresponding tokenized securities offering entity, wherein the voting contract includes a list of questions, and wherein a response to the list of questions by each holder of the tokenized securities is collected, processed, counted, and recorded as a vote on the immutable ledger; and wherein the response to the list of questions by each holder of the tokenized securities being recorded on the immutable ledger causes at least one action with regard to the tokenized securities to automatically execute.

2. The system of claim 1, wherein the at least one proxy contract includes a unique contract address, a unique address of the first party, a unique address of the second party, contract data which is being proxied, and/or terms to act on the at least one proxy contract, wherein the terms to act on the at least one proxy contract include a time the at least one proxy contract is signed, renewals, terminations, and methods to act on the contract data, and wherein the methods to act on the contract data comprise contract ownership transfer.
3. The system of claim 1, wherein the at least one proxy contract allows for ownership of the at least one proxy contract to be passed from the first party to a third party when specific contract terms are met.
4. The system of claim 1, wherein the securities token platform is configured to provide an aggregate contract between the first party and a multiplicity of parties, wherein the multiplicity of parties includes investors and parties included in proxy contracts, and wherein the aggregate contract allows the first party to make transactions on behalf of the multiplicity of parties and wherein the aggregate contract allows the first party to make transactions with the multiplicity of parties.
5. The system of claim 1, wherein the tokenized securities contracts include tokenized securities contracts deployed at different times, wherein common ownership hashes and/or keys link the tokenized securities contracts deployed at different times, wherein the tokenized securities contracts deployed at different times are configured to be transferred in one transaction.
6. The system of claim 1, wherein a term of the at least one proxy contract determines whether the first party or the second party is enabled to submit the response to the list of questions.
7. The system of claim 1, wherein the securities token platform is configured to provide at least one aggregate contract between the first party and a multiplicity of parties, wherein the first party represents the multiplicity of parties, wherein the voting contract inherits the at least one aggregate contract such that the voting contract is between the first party representing the multiplicity of parties and a corresponding tokenized securities offering entity, wherein terms of the voting contract are automatically applied to determine whether a simple majority, a super majority, or another calculation is applied to the responses to the list of questions to determine how the first party representing the multiplicity of parties for the at least one aggregate contract submits a vote.
8. The system of claim 1, wherein the securities token platform is configured to receive a request to trigger a transaction for a predetermined amount of fiat currency for the at least one securities token from a bank account.
9. The system of claim 1, wherein the tokenized securities offered by the tokenized securities offering entity represent common stock, preferred stock, options, bonds, notes, warrants, convertible notes, restricted stock units, and/or employee stock options, and wherein the server computer is configured to receive a return of tokenized securities upon payment of a restocking fee.
10. The system of claim 1, wherein the at least one proxy contract includes at least one smart contract, wherein terms of the at least one smart contract are automatically tracked and enforced.
11. The system of claim 1, wherein the at least one proxy contract is configured to be suspended,

resumed, and/or terminated by the tokenized securities offering entity.

12. The system of claim 1, wherein the securities token platform permits regulatory entities to prevent and/or allow the transactions of the tokenized securities.

13. A method for performing transactions of tokenized securities during a tokenized securities offering, comprising: providing a server computer for a tokenized securities offering entity in network communication with a securities token platform; the securities token platform recording tokenized securities contracts for the transactions of the tokenized securities; wherein the tokenized securities contracts comprise at least one proxy contract configured for contract ownership transfer from a first unique identifier to a second unique identifier, wherein the at least one proxy contract is recorded on an immutable ledger; a first party representing a second party for participating on the securities token platform via the at least one proxy contract; the securities token platform automatically maintaining a whitelist of accredited investors, wherein the whitelist is recorded on the immutable ledger; the securities token platform confirming accreditation credentials of the first party using a blockchain oracle, wherein the blockchain oracle is in real-time network communication with a plurality of data sources; the securities token platform automatically adding the first party to the whitelist upon confirmation that the first party is allowed to represent the second party; the sever computer confirming accreditation status of the first party and the second party based on the whitelist; wherein the tokenized securities contracts include encrypted information related to an owner of the tokenized securities contracts, wherein the information is encrypted using a first unique identifier associated with the owner; wherein upon a change of ownership of the at least one proxy contract, the encrypted information is decrypted and re-encrypted using a second unique identifier associated with a new owner; wherein the tokenized securities contracts include a voting contract, wherein the voting contract is between each holder of the tokenized securities and a corresponding tokenized securities offering entity, wherein the voting contract includes a list of questions, and wherein a response to the list of questions by each holder of the tokenized securities is collected, processed, counted, and recorded as a vote on the immutable ledger; and wherein the response to the list of questions by each holder of the tokenized securities being recorded on the immutable ledger causes at least one action with regard to the tokenized securities to automatically execute.

14. The method of claim 13, further comprising the server computer sending a link to documents related to at least one securities token to at least one user device.

15. The method of claim 14, further comprising the securities token platform generating a hash value for the documents based on whether the documents are viewed and/or signed.

16. The method of claim 14, further comprising the securities token platform receiving confirmation through use of at least one watermark that the documents related to the at least one securities token have been downloaded by the at least one user device.

17. The method of claim 14, further comprising the server computer updating the documents during the tokenized securities offering, and the securities token platform receiving a selection from the at least one user device regarding whether to keep or reverse a completed purchase order for the at least one securities token.

18. The method of claim 17, wherein a term of the at least one proxy contract determines whether the first party or the second party is enabled to submit the selection.

19. A method for performing transactions of tokenized securities during a tokenized securities offering, comprising: a securities token platform recording tokenized securities contracts for the transactions of the tokenized securities, wherein the tokenized securities contracts include encrypted information related to an owner of the tokenized securities contracts, wherein the information is encrypted using a first unique identifier associated with the owner, wherein the owner represents a second party for participating on the securities token platform via at least one of the tokenized securities contracts; the securities token platform automatically maintaining a whitelist of accredited investors, wherein the whitelist is recorded on an immutable ledger, wherein

the server computer is operable to confirm accreditation status of the owner and the second party based on the whitelist; the securities token platform confirming accreditation credentials of the owner using a blockchain oracle, wherein the blockchain oracle is in real-time network communication with a plurality of data sources, wherein upon confirmation that the owner is allowed to represent the second party, the securities token platform is configured to automatically add the owner to the whitelist; upon a change of ownership of the tokenized securities contracts, decrypting and re-encrypting the encrypted information using a second unique identifier associated with the second party, wherein the change of ownership is recorded on the immutable ledger; wherein the tokenized securities contracts include a voting contract, wherein the voting contract is between each holder of the tokenized securities and a corresponding tokenized securities offering entity, wherein the voting contract includes a list of questions, and wherein a response to the list of questions by each holder of the tokenized securities is collected, processed, counted, and recorded as a vote on the immutable ledger; and wherein the response to the list of questions by each holder of the tokenized securities being recorded on the immutable ledger causes at least one action with regard to the tokenized securities to automatically execute.
